



Externally set task 2016

Student Number: In figures

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In words

MATHEMATICS ESSENTIAL GENERAL COURSE

Students can use the following items for this task

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: non-programmable calculators approved for use in the externally set task
Notes on one A4 unfolded sheet to be collected at the end of the task

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Total time for the task: 50 minutes
Total marks: 33 marks
Weighting: 15% of the school mark

A formula sheet for use in this task is provided on page 11.

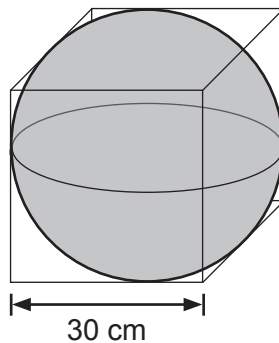
Question 1

(7 marks)

A ball manufacturer makes and delivers balls to various retail shops. The balls are inflated and individually packaged before being boxed for delivery.

The following diagram shows the ball in its individual packaging.

Packaging A



For each ball, determine:

- (a) the volume of the ball to the nearest cm^3 . (4 marks)

- (b) the amount of air, to the nearest litre (L) that would be required to inflate (fill) the ball. (2 marks)

- (c) To avoid damage during delivery, the empty spaces between the ball and its packaging needs to be filled with packaging pellets. Given the volume of the box is $27\,000\text{ cm}^3$, determine the volume of space that needs to be filled. (1 mark)

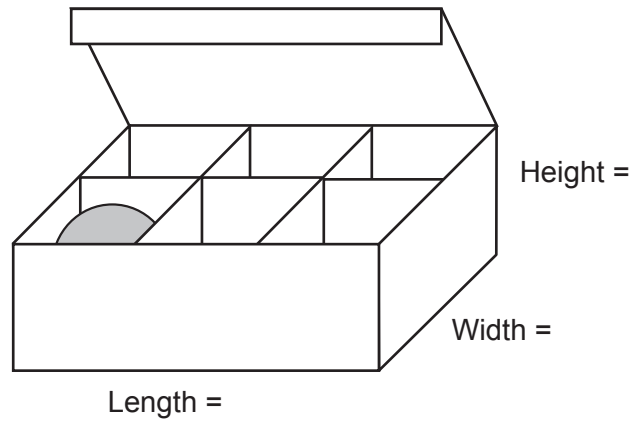
Question 2

(3 marks)

For delivery purposes, the company packs **six** balls (in their packaging) into boxes.

Insert the correct dimensions, in cm, for the length, width and height of each of these boxes.

Box A

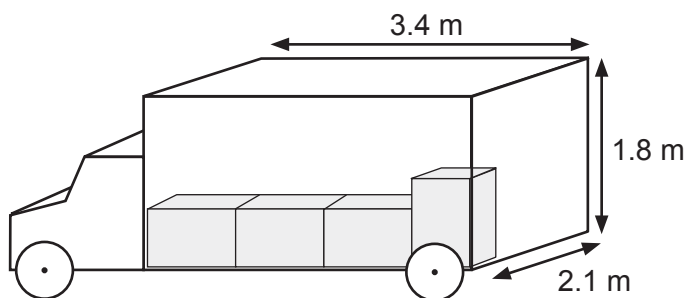


Question 3

(9 marks)

The following diagram shows the dimensions of the delivery truck.

The truck driver begins loading the truck with Box As for delivery, as shown in the diagram below.



If the driver continues to load the truck until he has placed the maximum number of boxes into the truck:

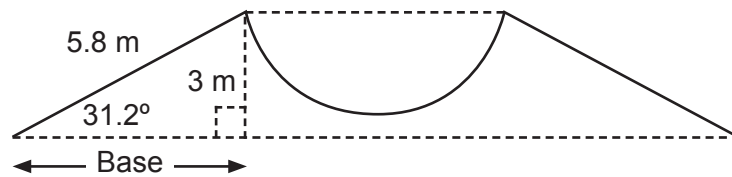
- (a) sketch the **side**, **top** and **back** views of the truck, showing the dimensions of the boxes inside the truck. (7 marks)

- (b) determine the maximum number of Box As that could fit into the truck. Show your working clearly to justify your answer. (2 marks)

Question 4

(14 marks)

Below is a cross-section of a skateboard half-pipe with ramps.



The designer calculated the base of the left ramp to be 6.5 m.

$$b^2 = 3^2 + 5.8^2$$

$$b^2 = 9 + 33.64$$

$$b^2 = 42.64$$

$$b = 6.5 \text{ m}$$

(a) (i) Why is this answer impossible?

(1 mark)

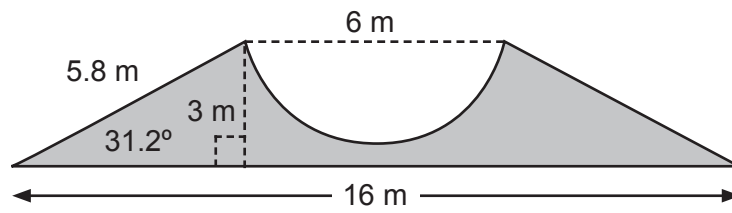
(ii) Determine the length of the base.

(4 marks)

- (b) The design involves a half-circle with a diameter of 6 m.

(3 marks)

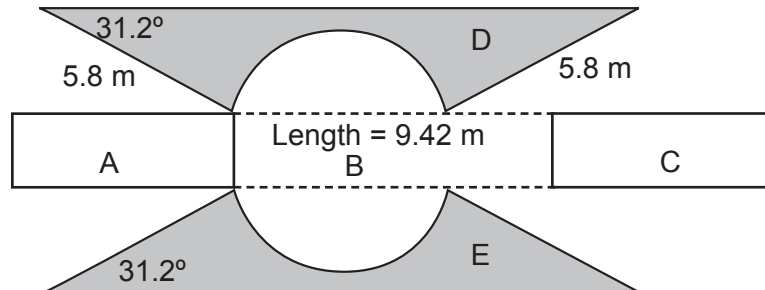
Show that the area of the cross-section is 18.9 m^2 .



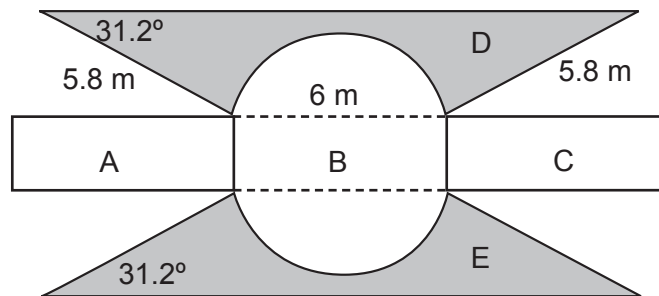
The ramp has a width of 3 m.

The designer produced two nets for the design of the ramp.

Net 1



Net 2



- (c) (i) Which of these two nets would you use to construct the ramp? (1 mark)
- (ii) Justify your choice mathematically. (1 mark)

- (d) Using your answer in parts (a) and (b), find the surface area of the skateboard half-pipe and ramps. (4 marks)

Space and measurement

In a right triangle: $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$ $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$ $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

Pythagoras' Theorem: In a right triangle ABC , where a , b are the short sides and c is the hypotenuse, $c^2 = a^2 + b^2$

Circle: $C = 2\pi r = \pi D$, where C is the circumference, r is the radius and D is the diameter

$A = \pi r^2$, where A is the area

Triangle: $A = \frac{1}{2}bh$, where b is the base and h is the perpendicular height

Parallelogram: $A = bh$

Trapezium: $A = \frac{1}{2}(a + b)h$, where a and b are the lengths of the parallel sides

Prism: $V = Ah$, where V is the volume and A is the area of the base

Cylinder: $S = 2\pi rh + 2\pi r^2$, where S is the total surface area
 $V = \pi r^2 h$

Sphere: $S = 4\pi r^2$
 $V = \frac{4}{3}\pi r^3$



Note: this marking key contains an amendment(s) post the delivery of the original marking key to schools. Any amendment is based on feedback received by the Authority during the administration and marking by schools of the Externally set task and is prior to the reviewer marking by the Authority.

MATHEMATICS ESSENTIAL

GENERAL COURSE

2016 Marking key for Externally set task

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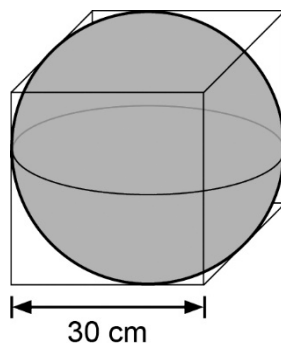
Question 1

(7 marks)

A ball manufacturer makes and delivers balls to various retail shops. The balls are inflated and individually packaged before being boxed for delivery.

The following diagram shows the ball in its individual packaging.

Packaging A



For each ball, determine:

- (a) the volume of the ball to the nearest cm^3 .

(4 marks)

Solution	
Radius of the sphere is 15 cm	
$V = \frac{4}{3}\pi r^3$	
$V = \frac{4}{3}\pi \times 15^3$	
$V = 14137.16694$	
$V = 14137 \text{ cm}^3$	
Specific behaviours	Marks
Chooses the correct formula to calculate the volume of the sphere	1
Substitutes the correct value for the radius	1
Correctly calculates the volume of the sphere	1
Rounds the answer to the nearest cm^3	1
Total	4

- (b) the amount of air, to the nearest litre (L) that would be required to inflate (fill) the ball.

(2 marks)

Solution	
$V = 14137 \text{ cm}^3$	
$= 14137 \text{ ml}$	
$= 14.137 \text{ L}$	
$= 14 \text{ L}$	
Specific behaviours	Marks
Correctly expresses the amount of air in ml or L	1
Correctly calculates the amount of air to the nearest L	1
Total	2

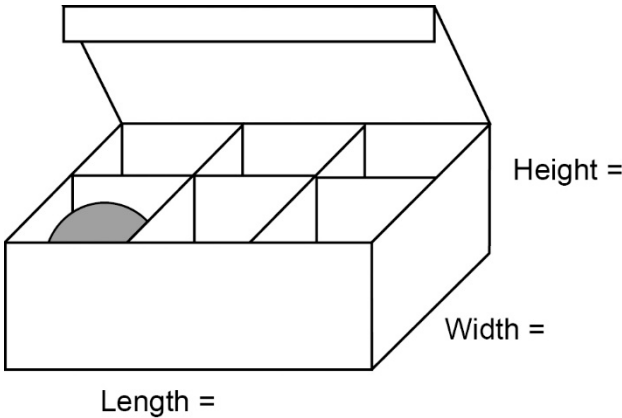
- (c) To avoid damage during delivery, the empty spaces between the ball and its packaging needs to be filled with packaging pellets. Given the volume of the box is 27 000 cm³, determine the volume of space that needs to be filled. (1 mark)

Solution	
$V = 27000 - 14137$ $V = 12863 \text{ cm}^3$	
Specific behaviours	Marks
Correctly determines the volume of space that needs to be filled	1
Total	1

Question 2 (3 marks)

For delivery purposes, the company packs **six** balls (in their packaging) into boxes.
Insert the correct dimensions, in cm, for the length, width and height of each of these boxes.

Box A



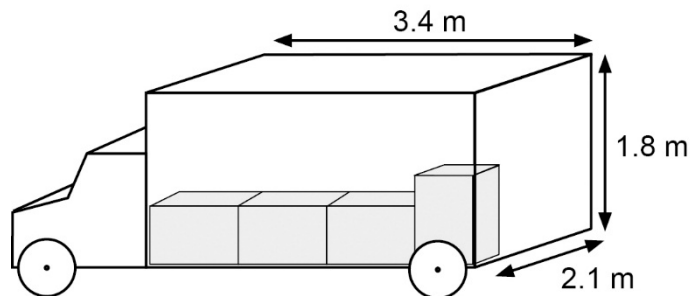
Solution	
Width = $2 \times 30 = 60$ cm	
Length = $3 \times 30 = 90$ cm	
Height = 30 cm	
Specific behaviours	Marks
Correctly calculates the width	1
Correctly calculates the length	1
Correctly labels the height	1
Total	3

Question 3

(9 marks)

The following diagram shows the dimensions of the delivery truck.

The truck driver begins loading the truck with Box As for delivery, as shown in the diagram below.



If the truck driver continues to load the truck until he has placed the maximum number of boxes into the truck:

- (a) sketch the **side**, **top** and **back** views of the truck, showing the dimensions of the boxes inside the truck. (7 marks)

Solution

Side view

90 cm 90 cm 90 cm 60 cm

60 cm 90 cm

3.4 m

Back view

30 cm

90 cm 90 cm

Top view

90 cm 60 cm

30 cm

Specific behaviours	Marks
Links dimension of box to side of truck	1
Links dimension of box to back of truck	1
Links dimensions of box to top of truck	1
Sketches correct side view	1
Sketches correct back view	1
Sketches correct top view	1
Labels views with dimensions	1
Total	7

- (b) determine the maximum number of Box As that could fit into the truck. Show your working clearly to justify your answer. (2 marks)

Solution

Side view

Back view

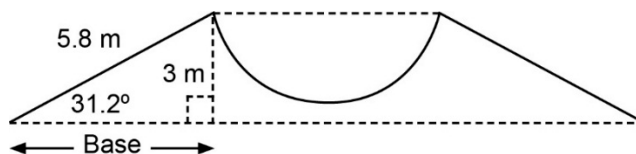
Top view

Specific behaviours	Marks
Determines the maximum number of boxes = 77	1
Provides evidence of method to determine the maximum number of boxes (includes volume of truck ÷ volume of a box then $12.852 \div 0.162 = 79.33$, therefore 79 boxes)	1
Total	2

Question 4

(14 marks)

Below is a cross-section of a skateboard half-pipe with ramps.



- (a) The designer calculated the base of the left ramp to be 6.5 m.

$$b^2 = 3^2 + 5.8^2$$

$$b^2 = 9 + 33.64$$

$$b^2 = 42.64$$

$$b = 6.5 \text{ m}$$

- (i) Why is this answer impossible? (1 mark)

Solution	
Sides of a right triangle must always be less than the hypotenuse	
Specific behaviours	Marks
States base of triangle must be less than hypotenuse in a right triangle	1
Total	1

- (ii) Determine the length of the base. (4 marks)

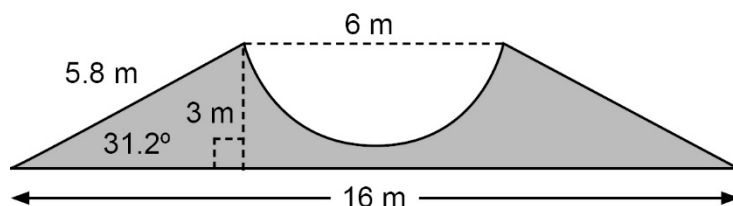
Solution				
Sides of a right triangle must always be less than the hypotenuse				
$5.8^2 = b^2 + 3^2$ $33.64 = b^2 + 9$ $24.64 = b^2$ $b = 4.96$ $b = 5.0 \text{ m}$	or	$\cos 31.2^\circ = \frac{b}{5.8}$ $b = 5.8 \times \cos 31.2^\circ$ $b = 4.96$ $b = 5.0 \text{ m}$	or	$\tan 31.2^\circ = \frac{3}{b}$ $b = \frac{3}{\tan 31.2^\circ}$ $b = 4.95$ $b = 5.0 \text{ m}$
Or, a scale diagram.				
Specific behaviours				Marks
States base of triangle must be less than hypotenuse in a right triangle				1
Uses an appropriate method				1
Correctly calculates the length of the base				2
Correctly expresses length of base to the nearest one decimal place				1
Total				4

- Note:** (1) The first behaviour 'States base of triangle must be less than hypotenuse in a right triangle' is not intended for the question and is therefore removed.
 (2) The behaviour 'correctly calculates the length of the base' should read 2 marks.

- (b) The design involves a half-circle with a diameter of 6 m.

Show that the area of the cross-section is 18.9 m^2 .

(3 marks)

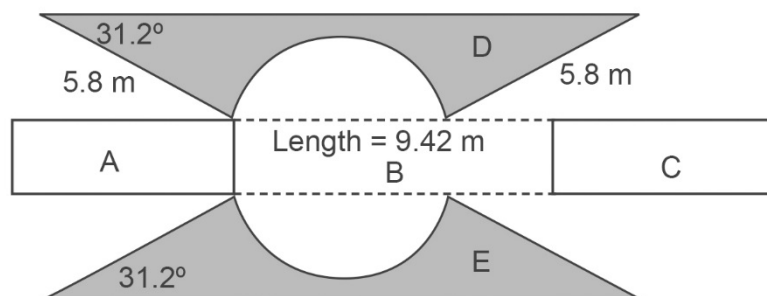


Solution	
$\text{Area} = \frac{1}{2} \times (6 + 16) \times 3 - \frac{1}{2} \times \pi \times 3^2$	
$\text{Area} = 33 - 14.14$	
$\text{Area} = 18.86 \text{ m}^2$	
$\text{Area} = 18.9 \text{ m}^2$	
Specific behaviours	Marks
Calculates area of trapezium (or composite shape)	1
Correctly calculates area of half-circle	1
Correctly calculates cross-section area	1
Total	3

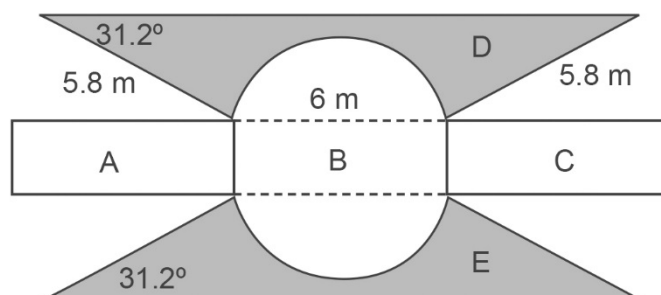
- (c) The ramp has a width of 3 m.

The designer produced two nets for the design of the ramp.

Net 1



Net 2



- (i) Which of these two nets would you use to construct the ramp? (1 mark)

Solution	
Net 1 – Length is equal to half the circumference of the circle	
Specific behaviours	Marks
Correctly recognises Net 1 as net required	1
Total	1

- (ii) Justify your choice mathematically. (1 mark)

Solution	
$\text{Length} = 3 \times \pi$ $= 9.42 \text{ m}$	
Specific behaviours	Marks
Justifies answer with valid reason	1
Total	1

- (d) Using your answer in parts (b) and (c), find the surface area of the skateboard half-pipe and ramps. (4 marks)

Solution	
$\text{Surface area} = 2 \times 18.9 + 2 \times 5.8 \times 3 + 9.42 \times 3$ $= 100.8 \text{ m}^2 \text{ (Accept 100.8 if 18.86 used)}$ or $\text{Surface area} = 2 \times 18.9 + 2 \times 5.8 \times 3 + 6 \times 3$ $= 90.6 \text{ m}^2 \text{ (Accept 90.5 if 18.86 used)}$	
Specific behaviours	Marks
Calculates the area of both sides	1
Calculates area of both ramps	1
Calculates area of curved section (half-pipe)	1
Calculates surface area	1
Total	4

